

Nishant Patel

Third Year Undergraduate | Department of Electrical Engineering
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Academic Qualifications

Year	Qualification	Institute	CPI/%
2021 - Present	B.Tech (EE)	Indian Institute of Technology Kanpur	8.5/10
2020	Class XII (CBSE)	Blooming Dale School, Budaun, UP	97.6%
2018	Class X (CBSE)	Blooming Dale School, Budaun, UP	95.0%

Scholastic Achievements

- Received the **Academic Excellence Award** *twice* for exceptional academic performance in 2021 and 2022 academic session
- Secured **99.7 percentile** and **All India Rank 3327** in **Joint Entrance Exam Mains 2021** amongst 1.3 million candidates
- Awarded **Merit Certificate** for scoring Full Marks in Social Studies in class X, was in **top 0.1%** among **1.6 Mn candidates**

Research Project

- Decentralised drone SWARMS and advanced UAV technology** 🦇 (May'23 - Jul'23)
Mentor: Prof. Ketan Rajawat, Department of Electrical Engineering, IIT Kanpur

Objective	● To increase the efficiency and usefulness of SWARM drones by addressing real life physical challenges
Approach	● Designed and implemented collision avoidance and congestion control algorithms with 88.4% accuracy ● Optimized the setup of SWARM using components like Raspberry Pi, Alfa Tube, GPS and others ● Examined all the technical requirements and specifications for each component, developed a plan to organize wiring, mount parts, and optimize system interaction , thereby enhancing the design of SWARM drones
Outcome	● With 81.2% accuracy , successfully carried out Leader-Follower tests for SWARM drones in group of 4

Key Projects

- Financial Analysis of NIFTY 50 Companies** (May'23 - ongoing)
Mentor: Prof. Suman Saurabh, Department of Industrial and Management Engineering, IIT Kanpur

Objective	● Data driven Analysis to compare and choose companies on the basis of financial ratios
Approach	● Developed an interactive NIFTY 50 firms R-Shiny dashboard , utilizing company financial statements ● Designed R code to calculate financial ratios using scraped data and generate comparative plots for analysis ● Enhanced dashboard for users to visualize market response using company information and output plots.
Outcome	●With 95%+ accuracy , users can compare or judge the financial health of stock(s) for given years

- Business Analytics Using Python | Summer Project** 🐍 | *Finance Analytics Club, IIT Kanpur* (Jun'22 - Aug'22)

Objective	● To extract max value from a firm's data assets to help them give insights that yield better, faster decisions.
Approach	● Learnt concepts of data analytics and manipulation, its's representation, Statistics and Machine learning. ● Created different models for price optimization and demand forecasting in Python using Price elasticity
Outcome	●Collaborating with an IOT startup Mobitra to clean and analyse data,identify trends for fleet owners

- Optimizing Travel Itineraries With Machine Learning** 🤖 | *Self Project* (Jun'23 - Jul'23)

Objective	● To optimize delivery vehicle travel routes using Machine Learning predictions for improved efficiency
Approach	● Utilized XGBoost model for complex numeric and categorical features, working over 120M+ data points ● Experimented with LightGBM model for faster processing of categorical data with weather inputs ● Implemented Genetic algorithm for taking predictions and construct time-efficient trip itinerary
Outcome	●With 93.2%+ accuracy , XGBoost model estimated a single trip's duration with error of 4.8 minutes only

Positions of Responsibility

- Assistant Placement Coordinator, Student's Placement Office, IIT Kanpur** (Apr'23 - Present)
*Previously worked as **Company Coordinator*** (Jun'22 - Mar'23)

Leadership	● Coordinating and Facilitating placement process for 1400+ students and internship process 700+ students. ● Spearheading a team of 80+ members handpicked responsibly for coordinating the internship drive. ● Developing corporate liaisons with 50+ corporate giants for Internship and Placement drive 2023-2024 .
Initiative	● Invited Startups for recruitment to increase success rate of internships,placements and enhance startup culture. ● Managed a unified database of 4500+ companies along with placement and internship portal of all students ● Managed Samvardhan career fest , coordinating participation of 15+ companies to facilitate networking
Impact	● Recorded highest number of offers(170+) made till date, on Day 1 of campus internship drive 2022-2023 ● Attuned Recruitment drive, securing 1200+ placement and 450+ internship offers as company coordinator

Relevant Courses

Data Structures and Algorithm*	Probability and Statistics	Fundamentals of Computing
Signals,Systems and Networks	Control Systems Analysis	Image Processing**

Technical Skills

Programming Languages	Libraries and Utilities
C/C++, ROS, HTML ,Python ,LaTeX, R	AutoCAD, Git, Gazebo, NumPy, Pandas, Matplotlib

Extra Curricular

Leadership	● Secretary, Games and Sports council - Managed Media and Coverage handles targetting 3K+ people ● Secretary, Photography Club, IITK- Organised and participated in various events and competitions ● Head Boy- Effectively managed and represented a student body of 3000 and enhanced successful school fests
Cultural	● Participated in Photostory competition and secured 3rd position among 22 IITs in Inter-IIT Cult Meet'23 ● Managed 300+ juniors who participated in Galaxy'23 (Inter-Hall Cult Championship), as Hall Captain